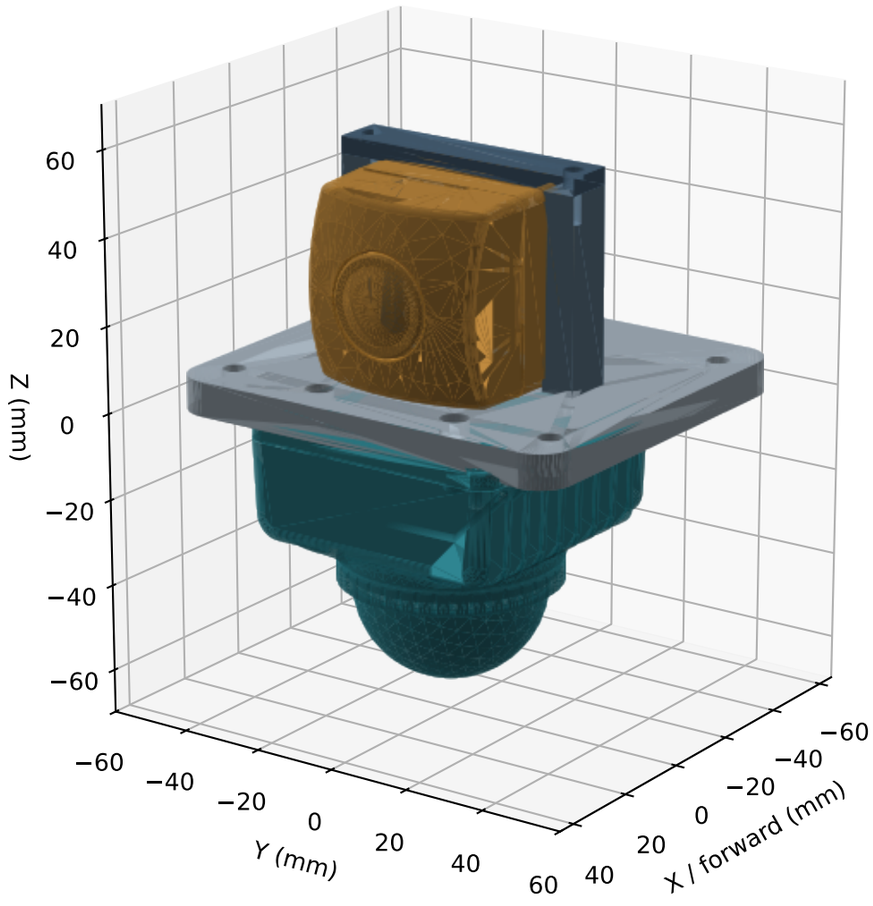
C2-176DEG + MID-360 | Vertical stack

CAMERA ABOVE / LiDAR FACING DOWN / BOTH CABLE PORTS TOWARD -X (REAR)

Actual user STEP geometry



SIDE | bounding outlines, mm

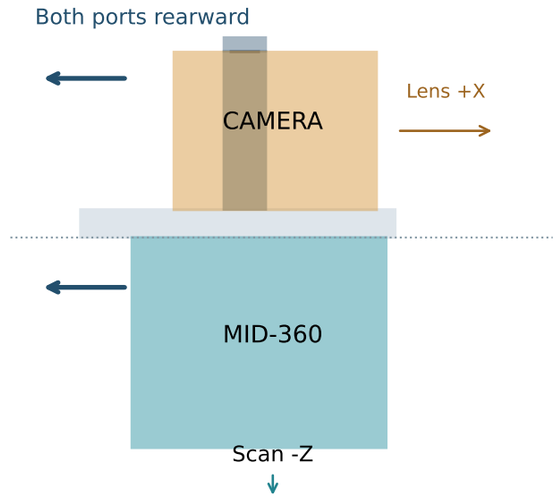
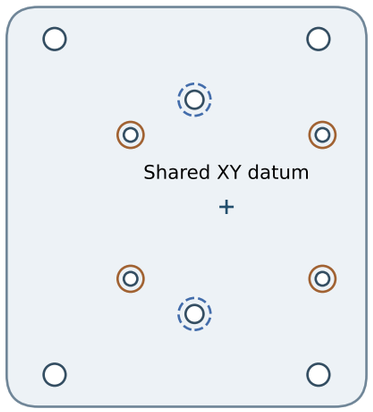


PLATE | top view, dashed = bottom counterbore



90 x 100 x 8 mm

Camera rests on plate TOP (Z=8). Inverted LiDAR mounts against plate BOTTOM (Z=0).

Review: screw engagement, clamp pressure, mating cables, field of view and vibration. Fasteners not included in CAD.

VERTICAL FIXTURE | Dimensions and assembly order (mm)

Axes: camera forward +X; camera cable -X; inverted LiDAR scan -Z, cable -X.
Plate: 90(X) x 100(Y) x 8, center (-10,0), R8 corners.
Camera bottom Z=8; LiDAR upper mounting face Z=0; common sensor XY datum=(0,0).

PLATE HOLES:
 HOST (-43, -42) dia 5.5 THRU
 HOST (23, -42) dia 5.5 THRU
 HOST (-43,   42) dia 5.5 THRU
 HOST (23,   42) dia 5.5 THRU
 LIDAR (-24, -18) dia 3.4 THRU; counterbore dia 6.5 x 3.5 deep from TOP
 LIDAR (-24,   18) dia 3.4 THRU; counterbore dia 6.5 x 3.5 deep from TOP
 LIDAR (24, -18) dia 3.4 THRU; counterbore dia 6.5 x 3.5 deep from TOP
 LIDAR (24,   18) dia 3.4 THRU; counterbore dia 6.5 x 3.5 deep from TOP
 POST (-8,-26.8) dia 4.5 THRU; counterbore dia 8 x 4.5 deep from BOTTOM
 POST (-8, 26.8) dia 4.5 THRU; counterbore dia 8 x 4.5 deep from BOTTOM

Posts x2: 12 x 8 x 45.2; center (-8,+/-26.8).
 Both ends: dia 3.3 drill depth 8; M4 effective thread depth 6 proposed.
Cap: 12 x 61.6 x 4; 2 x dia 4.5, pitch 53.6.
 Cap underside Z=53.2; camera top Z=53; top pad 8 x 30 x 0.2 compressed.
 Source: camera rotZ=-90; LiDAR rotX=-90 then rotZ=180 (degrees).
 MID-360 mounting centers from STEP: original X=+/-24, Z=+/-18.
 STEP bores suggest M3; exact thread and permissible insertion need confirmation.

ASSEMBLY ORDER (required because LiDAR covers lower post screw heads):
 1. Attach both posts with M4x8 from BOTTOM counterbores (4.5 mm engagement).
 2. Attach inverted LiDAR from TOP counterbores; M3x8 gives 3.5 mm insertion.
    M3x8 is conditional on manufacturer-approved insertion; confirm before use.
 3. Place camera; install top pad and cap with 2 x M4x8 (4 mm engagement).
 4. Attach host through 4 x dia 5.5 holes; M5 length depends on host.
 Lower post heads: dia 7 x 4; LiDAR heads: dia 5.5 x 3; both recessed 0.5.
 Tap geometry and fasteners not modeled. No bottom pads between sensors/plate.
 Deburr; support face flatness 0.05 and Ra3.2 proposed; validate requirements.
Metal mass at 2700 kg/m3: 0.218 kg.
 REVIEW: clamp friction/pressure, inverted operation, plugs/bends, FOV, loads.